Beyond Retrieval: Scope-Aware Memory Application in AI Agents

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Abstract

Long-term memory enables language-model agents to retain user preferences across interactions, but retrieving a relevant memory does not guarantee that the agent applies it in the correct context. We investigate scope-aware memory application by augmenting Mem0 and Supermemory with write-time and read-time scope tagging. We construct a benchmark of 40 episodes in 20 paired scenarios, covering five types of conflicting instructions, and evaluate eight conditions across 320 runs. The expected memory was retrieved in 43 of 51 incorrect responses across the six memory-system conditions, demonstrating a substantial retrieval–application gap. Read-time tagging with Supermemory achieves the highest observed adherence (92.5%), compared with 85% for its baseline, while write-time tagging reduces adherence for both systems. These findings suggest that explicit scope information can support memory application, but its effectiveness depends on how it is integrated and used.

1 Introduction

1.1 Motivation and Research Problem

Memory-augmented language-model agents retain user instructions across conversations, allowing responses to reflect previously expressed preferences. However, preferences are not always universally applicable. A user may generally prefer Polars for large datasets while requiring pandas for a particular group project. Both memories remain valid, but the appropriate action depends on the current context.

Existing memory systems primarily address the storage and retrieval of relevant information. Yet even when conflicting instructions are successfully retrieved, an agent must determine which instruction applies, whether an exception overrides a standing preference, and whether a temporary instruction should continue to influence subsequent tasks.

We investigate this distinction between *memory retrieval* and *memory application*: does an agent correctly use a retrieved instruction within its intended scope, and can explicit scope information improve its decisions?

1.2 Research Objectives and Contributions

We investigate two questions: (1) whether agents make incorrect decisions despite retrieving the expected memory, and (2) whether scope-aware interventions improve contextual adherence while preserving unrelated preferences.

Our contributions are:

1. A lightweight scope-tagging layer supporting standing, project-specific, and single-task instructions, integrated with Mem0 and Supermemory at write time or read time.
2. A synthetic benchmark of 20 paired scenarios (40 episodes) covering five memory-conflict categories.
3. An evaluation of eight conditions across 320 runs, separating retrieval success, memory application, scope errors, tagging accuracy, and unrelated-memory retention.

2 Related Work

2.1 Long-Term Memory in LLM Agents

Mem0 [1] provides persistent agent memory through extraction, storage, updating, and retrieval of information from interactions. Supermemory [4] offers a hosted memory service with ingestion and semantic search. Both aim to make relevant prior information available to agents without repeatedly supplying complete conversation histories.

Our work uses these systems as memory backends rather than proposing another storage or retrieval mechanism. We evaluate whether their retrieved information supports contextually appropriate decisions.

2.2 Memory Retrieval and Contextual Applicability

Retrieval relevance and contextual applicability are distinct. A memory may be relevant to the current topic but inapplicable to the current project, task, or condition. Conversely, a retrieved exception may need to override a broader preference without permanently replacing it.

We focus on this downstream decision problem by representing scope explicitly and evaluating both correct application and incorrect application beyond a memory's intended context.

3 Methodology

3.1 Scope Representation and Conflict Resolution

Each instruction is represented by three fields: *scope*, *key*, and *condition*. The scope determines applicability, the optional key identifies a project or task, and the optional condition specifies an additional constraint.

Table 1: Scope representation.

Scope	Applicability	Example
Standing	General preference across tasks	Use Polars
Project	Restricted to a named project	Marlin: pandas
Single-task	Restricted to one task	This export: CSV

The reference decision policy first excludes memories whose project, task, or condition does not match the current request. Among applicable memories concerning the same decision, scope precedence is

$$\text{single-task} \succ \text{project} \succ \text{standing}.$$

Within one scope, an applicable conditional instruction takes precedence over an unconditional instruction; recency resolves remaining conflicts at equal specificity. Unrelated instructions do not compete, and the benchmark excludes unresolved ties.

These rules define the reference actions and are communicated to the agent. The scope layer does not deterministically select the winning memory.

3.2 Scope-Tagging Architecture

We augment Mem0 [1] and Supermemory [4] with an LLM-based scope tagger. Figure 1 compares the two integration strategies.

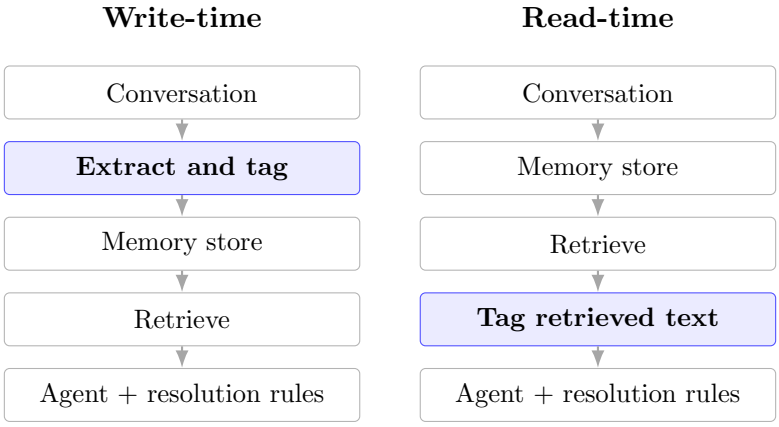


Figure 1: Scope-tagging integration strategies. Write-time tagging occurs before storage, whereas read-time tagging occurs after retrieval.

Write-time tagging extracts individual instructions from the full conversation and stores them with scope metadata. It can preserve scope cues early but may discard surrounding conversational context. Read-time tagging operates on the memory system's retrieved text, preserving native storage behaviour but relying on that text to retain sufficient contextual information.

Both strategies supply tagged memories and conflict-resolution rules to the agent. Structured task identifiers are withheld from the taggers, so task continuity must be inferred from natural-language context.

3.3 Benchmark Construction

3.3.1 Episode Structure and Mirrored-Pair Design

We construct 20 paired scenarios containing 40 episodes. Each pair includes two or three earlier conversation sessions, reference memories, and two probes sharing the same history. Probe A tests an applicable scoped instruction; Probe B changes the project, task, or condition so that the instruction no longer applies. Each episode consists of one probe and its shared history, with an expected action for evaluation.

Each pair also includes an unrelated standing preference and a retention check. Pair accuracy requires both probes to be correct, discouraging fixed choices that ignore context.

For example, pair `po-01` combines a standing preference for Polars on datasets exceeding one million rows with a project-specific instruction to use pandas for Project Marlin.

3.3.2 Conflict Categories and Dataset Generation

The benchmark covers five conflict categories, with four pairs per category:

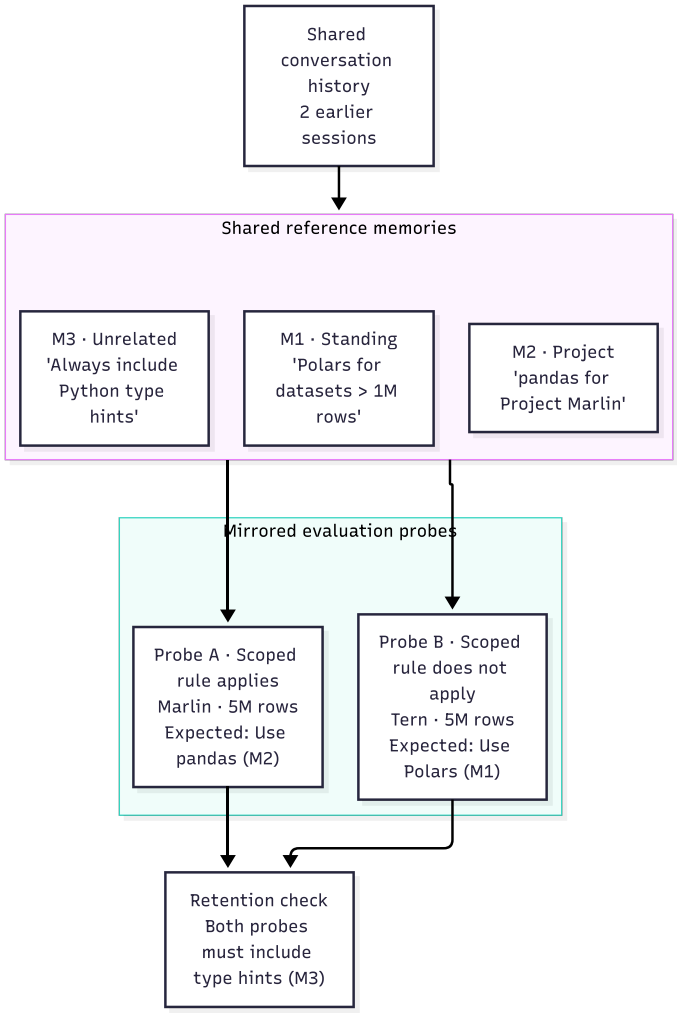


Figure 2: Mirrored-pair benchmark design illustrated using pair `po-01`. Both probes share the same conversation history and reference memories but differ in whether the project-specific instruction applies. An unrelated standing preference is evaluated independently in both probes.

Table 2: Benchmark composition.

Conflict category	Pairs	Episodes
Project override	4	8
Single-task override	4	8
Conditional scope	4	8
Same-scope supersession	4	8
Same-scope specificity	4	8
Total	20	40

The categories test project exceptions, temporary task exceptions, conditional applicability, replacement of older instructions within one scope, and precedence of more specific instructions within one scope.

The 20 pairs contain 66 distinct reference memories, including competing and unrelated instructions. Nineteen pairs were initially drafted with AI assistance and subsequently reviewed and revised. The suite varies domains, wording, chronology, and instruction positions; some standing preferences deliberately appear after narrower instructions.

3.3.3 Dataset Validation and Gold Labels

Automated checks validate the JSON schema, verbatim memory quotations, paired-probe similarity, expected decisions, and diversity. Only session and probe text reach the systems; structured gold labels and expected actions are withheld. A leakage test checks this separation.

The 66 memories were reviewed through two human annotation passes. After reconciliation, scope and key labels matched the original drafts. Four condition-expression differences were reviewed, and the original reference expressions were retained.

3.4 Experimental Conditions and Baselines

We evaluate eight conditions: no memory, full transcript, Mem0, Supermemory, and write-time/read-time scope-tagged variants of each memory system.

Table 3: Experimental conditions.

Condition	Memory supplied to agent
No memory	Current probe only
Full transcript	All earlier sessions
Mem0	Native retrieved memories
Supermemory	Native retrieved search results
Mem0 + write	Pre-extracted, tagged memories
Mem0 + read	Retrieved memories tagged afterward
Supermemory + write	Pre-extracted, tagged memories
Supermemory + read	Retrieved results tagged afterward

Each condition evaluates all 40 episodes, yielding 320 runs. The scope-layer conditions introduce tags, resolution rules, and memory ordering information together; their individual effects cannot be isolated.

4 Experimental Setup and Evaluation

4.1 Implementation and Model Configuration

The implementation uses Python 3.11+, `uv`, and a shared lockfile. The agent is `qwen/qwen3.8-flash` through OpenRouter, with temperature 0, seed 4213, a 1,024-token output limit, and reasoning disabled. The agent prompt template and sampling settings are fixed across conditions, with retrieval limited to ten results.

Qwen was selected after an operational pilot: it reported zero reasoning usage on 40 pilot calls, whereas DeepSeek returned empty text on three of eight calls when reasoning was disabled.

Both memory systems use a common interface and fresh namespaces for each episode and condition. Mem0 uses its default extraction and search, an `all-MiniLM-L6-v2` embedder (384

dimensions), Qdrant storage, and internal LLM calls through OpenRouter. The base `mem0ai` installation excludes optional spaCy and fastembed dependencies; the initial smoke test used vector retrieval without BM25. Supermemory uses its hosted API and waits for ingestion to complete before retrieval.

OpenRouter token usage and costs are recorded in an append-only ledger. Supermemory's separately billed processing is excluded from those costs.

4.2 Evaluation Metrics

An LLM judge identifies the action taken in each free-text answer. We evaluate the following metrics:

Table 4: Evaluation metrics.

Metric	Definition
Adherence	Expected action selected
Pair accuracy	Both A and B probes correct
Retrieval–application gap	Expected memory retrieved, wrong action selected
Retention	Unrelated standing preference followed
Tag accuracy	Predicted scope metadata matches reference
Failure stage	Earliest detected pipeline failure

Adherence is also separated into in-scope (A) and out-of-scope (B) probes. Incorrect decisions are classified as over-application, under-application, supersession, or other actions.

Retrieval is checked by verbatim matching or judge-assisted matching of rewritten notes. The full-transcript control counts as successful retrieval by construction; no memory counts as unsuccessful. Retention excludes unassessable answers.

Tagging accuracy is measured on notes associated with one reference memory. Scope is matched exactly; project keys must contain the reference project name, while single-task keys are excluded from scoring. Conditions use strict (variable, operator, value) and lenient (operator, value) matching.

Failure-stage categories are extraction, retrieval, scope loss, tagging, and reasoning. Each incorrect answer is assigned to the earliest detected failure. These are diagnostic classifications, not established causal mechanisms. Supermemory's internal extraction cannot be inspected separately from retrieval.

4.3 LLM-as-a-Judge and Statistical Analysis

The judge uses the same Qwen model at temperature 0. It receives the probe, response, and alphabetically ordered candidate actions, but not the expected action. It returns an action, `none`, or `mixed`, with supporting evidence, and separately evaluates the retention check.

The recorded human review covers 40 action labels and 38 assessable retention labels, with complete agreement against the judge (Cohen's $\kappa = 1.00$) [2]. The judge and agent share a model, so correlated errors remain possible; LLM-judge biases are discussed by Zheng et al. [6].

We report 95% Wilson confidence intervals [5] and exploratory exact McNemar tests [3] on the same episodes. Because the A/B probes within a pair share history, episode-level tests do not fully account for dependence; p-values are interpreted cautiously.

5 Results and Analysis

All 320 runs completed without execution errors.

5.1 Overall Correction Adherence

Figure 3 shows adherence across the eight conditions. No memory achieves 20% (8/40), versus 80% (32/40) with the full transcript. Mem0 and Supermemory achieve 77.5% (31/40) and 85% (34/40).

Read-time tagging achieves 80% with Mem0 and 92.5% (37/40) with Supermemory, the highest observed score. Write-time tagging reduces adherence to 60% and 77.5%, respectively. Supermemory + read-time resolves 17/20 complete pairs, compared with 15/20 for its baseline. Its improvement is not statistically significant under the reported McNemar test ($p = 0.25$).

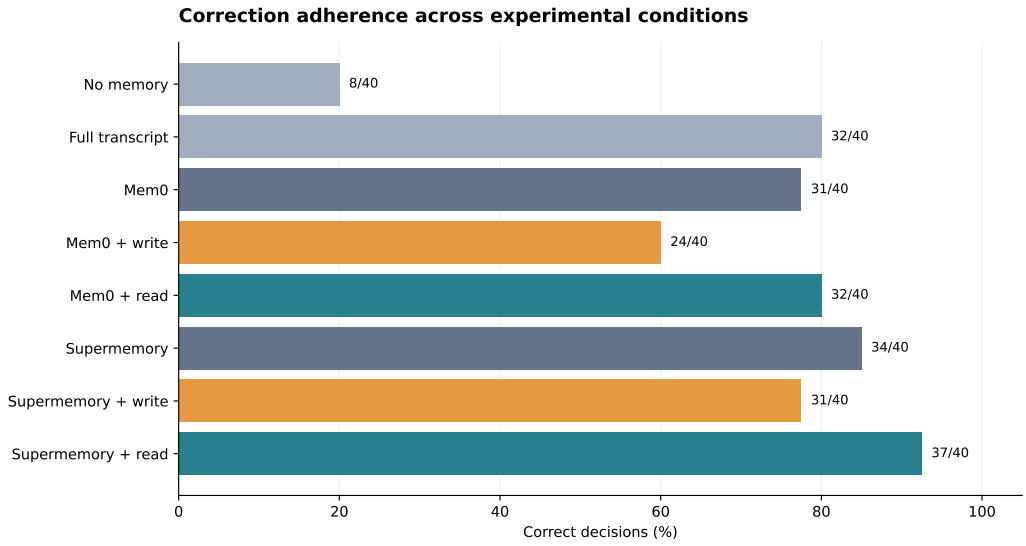


Figure 3: Correction adherence across eight conditions (40 episodes per condition).

5.2 Retrieval Versus Application

Across the six memory-system conditions, 43 of 51 incorrect responses occurred despite retrieval of the expected memory (Figure 4). Unmodified Mem0 and Supermemory retrieved the expected memory in all 40 episodes but made nine and six incorrect decisions, respectively. Read-time tagging reduced these counts to eight and three.

Thus, successful retrieval is insufficient for correct application, although a retrieved note may still have lost important scope information.

5.3 Scope-Boundary and Conflict-Type Performance

Figure 5 reveals asymmetric write-time errors. Mem0 + write correctly handles 10/20 in-scope probes, versus 17/20 for Mem0. Supermemory + write handles 11/20, versus 18/20 for its baseline, but correctly handles all 20 out-of-scope probes. This indicates substantial under-application of scoped instructions.

Supermemory + read-time handles 20/20 in-scope and 17/20 out-of-scope probes.

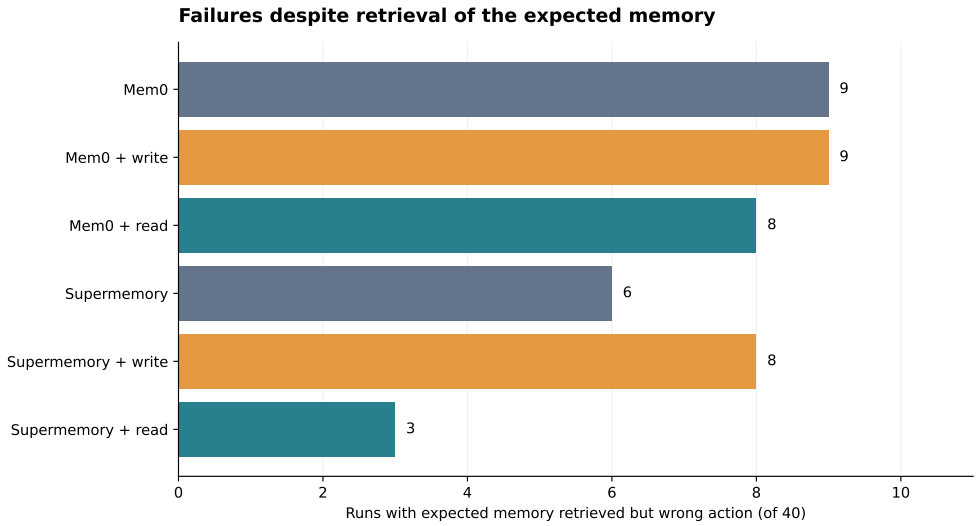


Figure 4: Incorrect actions despite retrieval of the expected memory.

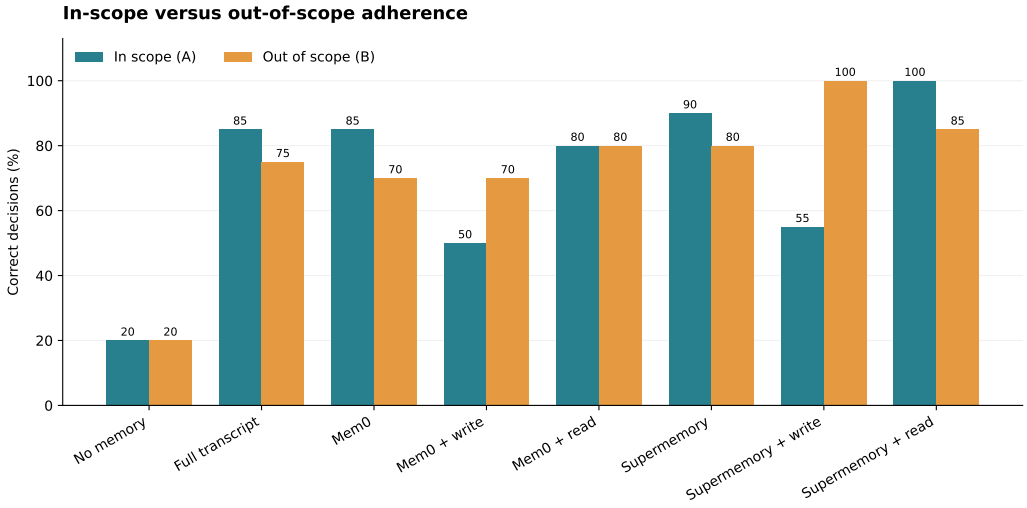


Figure 5: Adherence on in-scope (A) and out-of-scope (B) probes, 20 of each per condition.

Across conflict types (Figure 6), conditional scope and same-scope supersession are handled reliably by most memory-system conditions. Single-task overrides are hardest: Mem0 and Supermemory each achieve 4/8, while read-time Supermemory reaches 6/8. Same-scope specificity also remains challenging, with 4–7/8 correct among the full-transcript and memory conditions.

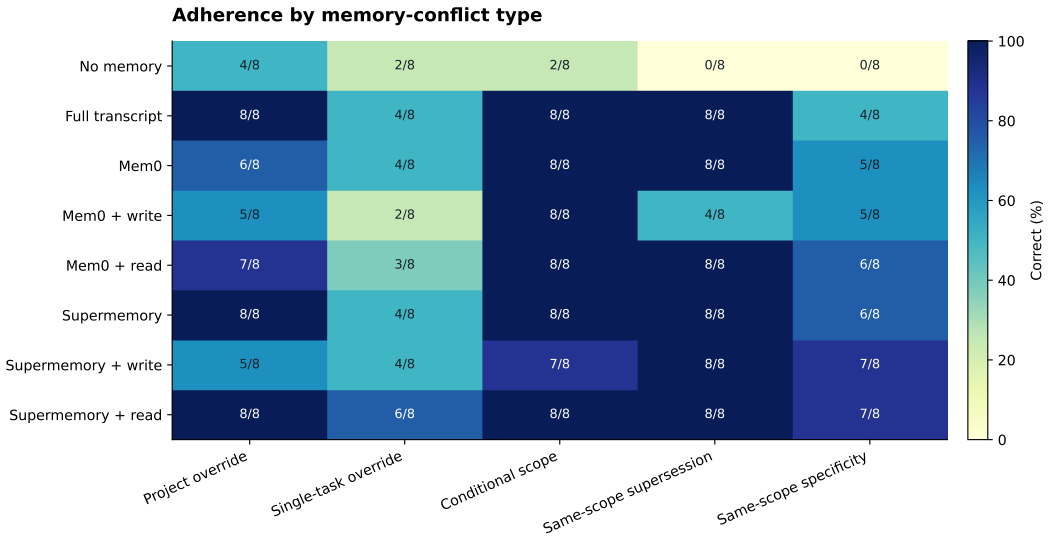


Figure 6: Adherence across five conflict categories; eight episodes per category.

5.4 Failure Diagnosis, Tagging, and Retention

The diagnostic breakdown in Figure 7 attributes most Supermemory baseline errors to reasoning despite available information. Mem0 exhibits five baseline scope-loss errors; write-time Mem0 has seven extraction and four tagging failures among its 16 incorrect responses. Supermemory + write-time has eight reasoning-classified failures among nine.

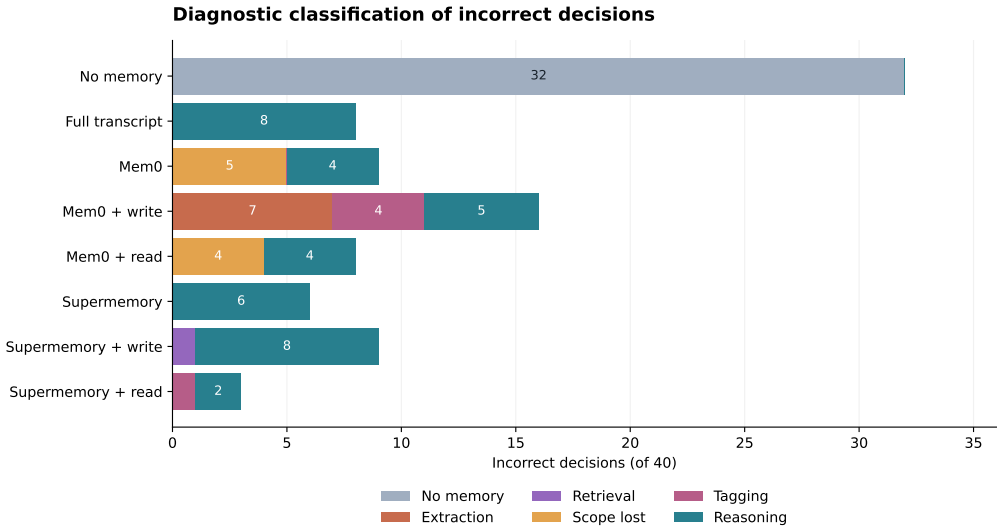


Figure 7: Incorrect decisions by earliest detected failure stage; classifications are heuristic.

Write-time tagging achieves 100% scope accuracy on evaluated tags for both systems, versus 87% for read-time Mem0 and 77% for read-time Supermemory. Nevertheless, the highest downstream adherence occurs with read-time Supermemory, illustrating that tag accuracy alone is insufficient. These figures use different eligible-note sets and are not directly controlled comparisons.

Unrelated-memory retention is 95% with the full transcript, 67.5% with Mem0, and 97.5% with Supermemory. Scope-aware Mem0 variants retain 62–72%, compared with 87–98% for

Supermemory variants.

6 Discussion and Limitations

6.1 Write-Time versus Read-Time Tagging

Read-time tagging matches or improves adherence relative to both memory-system baselines, whereas write-time tagging reduces it. One possible explanation is that write-time extraction isolates instructions from conversational context, making task continuity difficult to infer. Conversely, read-time tagging retains the underlying memory system's representation, although its tags may be less accurate.

Write-time Supermemory illustrates the trade-off: it avoids over-applying scoped memories on all 20 out-of-scope probes but fails to apply many scoped instructions when appropriate.

6.2 Why Retrieval and Tagging Are Insufficient

The retrieval–application gap shows that access to a relevant instruction does not ensure correct conflict resolution. In pair `po-01`, for example, an agent may select the standing Polars preference despite the applicable project-specific pandas instruction. Other failures involve carrying a single-task exception into a new task or combining conflicting instructions.

These cases suggest that effective memory requires both relevant information and reliable contextual decision-making. Representative examples are included in Appendix A.

6.3 Threats to Validity

The benchmark contains only 20 synthetic pairs with short histories and explicit scope cues. Nineteen pairs were initially AI-assisted, and numeric conditions avoid threshold boundaries. The findings may not generalize to longer, ambiguous, or naturally occurring interactions.

The study uses one agent model and one run per condition. Fixed sampling settings do not guarantee reproducibility, while Supermemory's hosted processing is not fully observable.

The scope treatments jointly change metadata, resolution instructions, and memory ordering, preventing attribution of observed effects to scope tags alone. Retrieval matching and failure-stage diagnoses were not independently validated in full. The gold labels underwent human–AI review rather than independent double-human adjudication.

Finally, the two probes within each pair are dependent, limiting the interpretation of episode-level significance tests.

7 Conclusion

We evaluated whether scope-aware memory handling helps agents apply retrieved instructions in the correct context. Across 320 runs, incorrect memory application frequently occurred even when the expected memory was retrieved.

Read-time tagging with Supermemory achieved the highest observed adherence, while write-time tagging reduced performance for both memory systems. The results suggest that scope-aware interventions can be useful, but explicit metadata alone does not resolve contextual memory conflicts. Larger benchmarks and controlled ablations are needed to isolate the effects of tagging, conflict-resolution instructions, and memory ordering.

Individual Contributions

Member	Contributions	Share
Isaac Chin Wei En	Benchmark and environment	–
Edward Chew Kian Boon	Scope layer and logging	–
Ng Cheng Kiat Nicholas	Evaluation and scoring	–
Jagan Mohan Shailesh	Validation, analysis, report	–

References

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A Qualitative Failure Cases

A.1 Project Override: `po-01`

The user generally prefers Polars for datasets over one million rows but requires pandas for Project Marlin. Probe A concerns Marlin and expects pandas; Probe B concerns Project Tern and expects Polars. The unrelated type-hint preference applies to both.

In the logged project-override failure, the agent selects Polars despite the applicable Marlin instruction. This illustrates a failure to prioritize project scope over a standing preference.

A.2 Single-Task Override: `st-04`

The user generally prefers inline SQL subqueries, requires CTEs for Project Marlin, and requests views for one particular task. Probe A expects views; Probe B expects CTEs. Some conditions carry the temporary exception into Probe B, while others fail to apply it in Probe A.

A.3 Same-Scope Specificity: `sp-02`

The user normally opens emails with "To whom it may concern," but prefers "Hey there," when writing to a neighbour. Probe A concerns a neighbour; Probe B concerns a landlord.

Some responses combine both greetings or apply the conditional greeting outside its intended scope.

B Supplementary Experimental Results

The full pair-by-pair adherence matrix, tagging-accuracy breakdown, failure-stage counts, and per-condition cost accounting are available in `reports/summary.md` and `reports/episode_scores.csv`.

OpenRouter spending for the full experiment was approximately $0.15, with approximately $0.06 for judging. Supermemory's separately billed processing is excluded.

C Reproducibility and Implementation Details

The implementation is available in the project repository. Experimental conditions are defined in `agent/conditions.py`; runs are executed through `agent/runner.py`; and scoring is implemented in `scoring/`.

The 320 existing run logs, cached judge verdicts, and cost ledger allow rescoring without repeating agent calls. Figures are generated from the scoring outputs in `reports/`.